PH PHEWPHONICS · MAD FRAMEWORK

MAD RUN — OCTOBER 2026 · AA · DA · SSA

Autodesk Vault Professional
On-Premise to Cloud Migration

Full Multi-Agent Debate output: Advocate (AA), Devil's Advocate (DA), and Synthesiser (SSA) verdict. Generated by the MAD Framework from runs/vault-cloud-migration/.

● CONDITIONAL GO

MAD FRAMEWORK | AA · DA · SSA | OCTOBER 2026 | VAULT-CLOUD-MIGRATION

SYNTHESISER VERDICT

SSA Output — Final Verdict

Synthesised from full AA and DA debate. This is the actionable output.

SSA OUTPUT: Synthesiser Verdict — Autodesk Vault Professional Cloud Migration

MAD Cycle | Topic: Vault On-Premise to Cloud Migration

Role: Synthesiser (SSA) — Final Verdict

Run date: October 2026

AA output: AA-output.md | **DA output:** DA-output.md

VERDICT: CONDITIONAL GO

The migration is technically viable and operationally justified — but not yet ready to execute.

Thirteen critical failure modes have been identified that standard IT migration testing will not catch. Nine of these are specific to Plant 3D and Inventor workflows. Until six named conditions are met, go-live carries unacceptable operational risk.

This is not a "stop and review" verdict. It is a structured set of conditions that, once met, make this migration safe to execute. The conditions are achievable within a defined preparation window.

WHAT BOTH SIDES AGREE ON

POINT	AGREED
Vault Professional runs legitimately on cloud VMs	Yes — Autodesk-supported, documented
The migration procedure exists and is understood	Yes — ADMS Console wizard, backup/restore
All file formats are technically preserved through migration	Yes — SQL + file store both migrate
Plant 3D check-in direction constraint exists	Yes — but AA and DA disagree on whether it becomes harder to get right post-migration
Latency affects assembly open performance	Yes — the disagreement is on whether AVFS mitigates it sufficiently
IT's standard test cases are insufficient for this environment	Yes — both agree on this

THE CORE DISPUTE: WHERE AA AND DA DIVERGE

Plant 3D UX Risk (1.1)

AA position: The check-in constraint is unchanged; cloud doesn't make it worse.

DA position: A working folder path change post-migration changes the visual cues users rely on. Engineers who have muscle memory for a specific flow will encounter a new path presentation and some will attempt the wrong workflow —

not from error, from a changed interface.

SSA finding: DA is correct on the mechanism; AA is correct that the constraint itself doesn't change. The risk is transitional — highest in the first four weeks post-go-live. Mitigation is explicit: user briefing + a tested "wrong workflow" scenario that confirms what actually happens.

Latency and Assembly Performance (2.1)

AA position: AVFS cache nodes resolve the latency problem — this is the canonical multi-site architecture.

DA position: AVFS has not been provisioned. At 60ms round-trip without AVFS, large assembly open times become operationally disruptive.

SSA finding: AA's AVFS answer is correct but it is future-tense. The current proposed migration does not include AVFS — it is a straight cloud lift. Without AVFS, the DA calculation is accurate. AVFS must be scoped into the architecture before go-live if latency measurements confirm it is needed.

THE SIX GO/NO-GO CONDITIONS

These are binary — each is either met or it is not. All six must be met before go-live.

CONDITION 1: Latency Measured and Within Threshold

What: Measure actual round-trip latency from every workstation through the corporate proxy to the cloud Vault server. Time the three largest assemblies opening with 10 concurrent users simulated.

Threshold for Go: Effective latency ≤ 40ms. If latency is 40–70ms: AVFS must be added to the architecture before go-live. If latency > 70ms: escalate to engineering leadership before proceeding — this is a blocker.

Why IT won't do this by default: Standard connectivity testing confirms a connection exists, not whether it performs adequately for CAD operations under load.

Owner: IT (measurement) + Engineering Lead (threshold sign-off)

CONDITION 2: Plant 3D Test Cases Written and Signed Off by a Plant 3D Operator

What: An engineer who uses Plant 3D daily — not IT, not a project manager — must review the full test case list and confirm these scenarios are covered:

- Check-in direction test (attempt the wrong workflow; document what breaks)
- Project.xml path resolution after working folder change
- Lock file behaviour with two users simultaneously editing the same drawing
- Isometric generation output landing in correct Vault location
- Data Manager connection to Plant 3D SQL instance from cloud environment
- Pipe spec catalog path resolution

Why this condition exists: DA finding 9.1 — test cases written without Plant 3D operator input structurally miss the failure modes that matter most. This is the meta-risk that makes all other risks harder to catch.

Owner: Named Plant 3D operator (sign-off) + IT (execution)

CONDITION 3: Version Compatibility Matrix Confirmed for Every Workstation

What: Document the exact Vault client add-in version on every workstation. Cross-reference against Autodesk's official compatibility matrix for the cloud Vault server version. Confirm match. Get confirmation in writing from the cloud provider or Autodesk VAR.

Why this condition exists: A version mismatch is a day-one outage for every user simultaneously. This is not a partial failure — it prevents all Vault access from Inventor, AutoCAD, and Plant 3D.

Owner: IT

CONDITION 4: Project.xml and Working Folder Path Remediation Plan Documented and Tested

What: Before migration:

1. Identify whether the Vault working folder path will change post-migration
2. If yes: identify every project.xml file in every Plant 3D project that references the old path or old server hostname
3. Document the exact remediation steps (which files to edit, what to change, who does it)
4. Execute this remediation in the test migration environment and confirm Plant 3D opens every project successfully

Why this condition exists: If project.xml files still reference the old server hostname post-go-live, every Plant 3D project opens in degraded mode. This is total loss of Plant 3D functionality.

Owner: IT (path change confirmation) + Plant 3D operator (project.xml audit)

CONDITION 5: Rollback Procedure Executed in Test Environment

What: The rollback plan must be run — not documented, not read, run — in a test environment before go-live. Confirm: (a) every file is recoverable from the pre-migration snapshot, (b) the maximum data loss window is defined and accepted in writing by engineering leadership, (c) the procedure for handling files that exist only in cloud Vault during the parallel run is defined.

Why this condition exists: An undocumented rollback is a recovery plan. An untested rollback is not. DA finding 8.1 — two weeks of data loss if rollback is required and the procedure has never been rehearsed.

Owner: IT (execution) + Engineering Manager (written sign-off on data loss window)

CONDITION 6: Checked-Out Files Cleared Before Migration

What: Audit every currently checked-out file across all users. Define the explicit procedure for each engineer to resolve their checked-out state. Make this a hard technical prerequisite — Vault migration begins only after all files are checked in.

Why this condition exists: Files checked out to the old working folder path at migration time are orphaned. Inventors with 200-part assemblies checked out mid-design cannot check back in without a complex manual resolution procedure that risks overwriting in-progress work.

Owner: Engineering Lead (communication) + IT (audit)

UNRESOLVED RISKS REGISTER

Issues that remain open after conditions are met and must be monitored post-go-live.

RISK	RATING	MONITORING
Lock file race under concurrent load	SIGNIFICANT	Track any Plant 3D data conflicts in first 4 weeks
iLogic Vault property reads on cloud	SIGNIFICANT	Test all iLogic rules in test environment before go-live

RISK	RATING	MONITORING
Sheet set (.dst) absolute path failures	SIGNIFICANT	Audit and test-plot all sheet sets before go-live
DWF/DWFx visualisation currency	MINOR	Spot-check web viewer after migration
Isogen output path accuracy	SIGNIFICANT	Run isogen on 5 known P&IDs post-migration; compare to reference
Excel BOM item number continuity	CRITICAL (if SQL rebuilt)	Export item master pre/post; compare row by row
Historical revision binary file integrity	CRITICAL	Script-test open of old revisions across 100 sample files
Plant 3D pipe spec catalog path	SIGNIFICANT	Test spec validation on restricted component placement
IFC parent-child link integrity	SIGNIFICANT	Verify 10 IFC output attachments post-migration

STRANGLER FIG CUTOVER PLAN

Recommended approach: **phased cutover by application, not by project.**

Week 0 (Preparation): All six conditions met. Communicate to all engineers. All files checked in. Version compatibility confirmed.

Week 1–2 (AutoCAD users): AutoCAD users cut over first. Lowest migration risk, fastest to verify xref resolution. AutoCAD Vault add-in is better-documented than Plant 3D. Any path issues surface here at lowest cost.

Week 3–4 (Inventor users): Inventor cut over. Latency monitoring active. Assembly open times tracked. iLogic rules tested. AVFS in place if latency tests required it.

Week 5–6 (Plant 3D — last): Plant 3D cut over last. All project.xml remediation complete. Plant 3D operator on site for day-one support. Lock file behaviour monitored actively.

On-premise Vault: Set to read-only immediately at each application's cutover, not at go-live of the full system. This prevents divergence without requiring simultaneous cutover of all users.

Decommission: 60 days after final Plant 3D cutover, with no open rollback issues, on-premise decommissioned.

ROLLBACK TRIGGERS

If any of the following occur, pause and escalate immediately:

TRIGGER	ACTION
Plant 3D check-in fails for any user post-go-live	Pause Plant 3D cutover; investigate path/hostname before proceeding
Assembly open time > 90 seconds for any user	Activate AVFS evaluation immediately; do not proceed with Inventor cutover until resolved
Any Vault add-in reports "cannot connect to server" from client machine	Full stop; verify SQL hostname config and version matrix

TRIGGER	ACTION
Any SQL consistency check error post-migration	Full stop; resolve before any user check-ins
Historical revision opens with "file not found"	Log immediately; do not proceed with decommission until all instances resolved
Two concurrent Plant 3D check-ins to same file both succeed	Treat as critical data integrity event; audit both versions

WHAT WAYNITO SHOULD BRING TO TESTING

You are walking into testing to find problems before go-live. Use the DA output as your testing checklist. The highest-value scenarios to test, in priority order:

1. **Plant 3D wrong-direction test:** Open a Plant 3D drawing from Vault Explorer (not from within Plant 3D). Edit it. Try to check in. Document exactly what happens.
1. **Concurrent Plant 3D editing:** Two users open the same drawing simultaneously. Both edit and check in. Does the second user see the first user's lock? What happens to the overwritten version?
1. **Large assembly open timing:** Open the three largest assemblies with 5–10 people simultaneously doing other Vault operations. Record the time from click to fully loaded.
1. **iLogic rule execution:** Find a drawing with an iLogic rule that reads a Vault property. Run the rule on the cloud system. Verify the returned value matches what Vault actually shows.
1. **Isometric generation path:** Run isogen on a P&ID. Confirm the output appears in Vault under the correct revision. Compare content to a known-good reference.
1. **Historical revision open:** Find a drawing with 5+ revisions. Open each old revision from Vault history. Confirm all open successfully.
1. **Sheet set batch plot:** Locate a sheet set (.dst) file. Run batch plot. Confirm all sheets plot successfully.
1. **Version matrix check (before testing begins):** Confirm the cloud Vault server version matches the Vault client add-in on your test machine. If not, stop and escalate before testing anything else.

SSA SUMMARY

QUESTION	ANSWER
Is this migration technically possible?	Yes
Is it officially supported by Autodesk?	Yes
Is it ready to execute right now?	No — six conditions not yet met
What is the highest single risk?	Plant 3D test cases not written by a Plant 3D operator (Condition 2)
What is the right sequencing?	AutoCAD → Inventor → Plant 3D (not simultaneous)
Is Plant 3D safe with cloud Vault?	Yes, with correct workflow — but the transition period requires active monitoring

QUESTION	ANSWER
Does AVFS resolve the latency problem?	Yes — if it is in scope. Current plan does not include it.
What should Waynito do in testing?	Run the 8 scenarios above; document every deviation from on-premise behaviour

Final position: Conditional Go. The migration should proceed — after the six conditions are met, in the application sequence defined above, with rollback triggers defined and plant 3D operator sign-off on test cases.

ADVOCATE AGENT

AA Output — Strongest Case FOR

Full advocate position. All arguments for proceeding.

AA OUTPUT: Advocate — Autodesk Vault Professional Cloud Migration

MAD Cycle | Topic: Vault On-Premise to Cloud Migration

Role: Advocate (AA) — Strongest Case FOR

Run date: October 2026

POSITION STATEMENT

A cloud-hosted Autodesk Vault Professional environment is the correct strategic direction for an engineering/drafting department operating with Inventor, AutoCAD Plant 3D, and AutoCAD. The migration is technically viable, operationally supported, and delivers concrete benefits that on-premise hardware cannot match — particularly around resilience, IT overhead reduction, and multi-site access. The DA will attack; this case is built to withstand it.

1. GENUINE OPERATIONAL BENEFITS

1a. Infrastructure Elasticity

On-premise Vault is frozen to the hardware purchased. Cloud storage expansion is a configuration change — no procurement, no downtime, no physical work. Compute can also be scaled vertically if Vault indexing or SQL query times degrade.

1b. Disaster Recovery and Backup Quality

Cloud providers offer native, automated backup services (AWS S3/Glacier, Azure equivalents) vastly superior to typical on-premise tape rotation or NAS backup. Point-in-time restore is achievable for a department where losing Inventor assemblies or issued drawing revisions is catastrophic.

1c. Reduced IT Dependency

On-premise Vault requires Windows Server licensing, SQL Server licensing, OS patch management, SQL maintenance plans, antivirus exclusion management, disk health monitoring, and UPS. In a managed cloud or IaaS model, most of this shifts to the provider. Small engineering departments with limited IT staff benefit directly.

1d. Global and Remote Access

Cloud-hosted Vault enables engineers working remotely or from a second office to access full Vault Professional functionality — check-in/check-out, lifecycle transitions, BOM queries — without a VPN hairpin back to a physical server room.

1e. Multi-Site Replication (AVFS)

Vault Professional's AVFS replication component allows on-premise replica nodes at each office with the Vault Server in the cloud. This is the canonical multi-site Vault architecture and works better with a cloud-hosted central server than a single on-premise server that gives remote sites no replica at all.

2. AUTODESK'S OFFICIAL SUPPORTED PATHWAY

Autodesk explicitly supports hosting Vault Professional on cloud infrastructure. Documented in AU class MFG229048 ("SkyVault"):

- Vault Workgroup and Vault Professional may be virtualised, subject to licensing
- Both AWS and Azure are supported hosting environments
- Supported topologies: single-server VM, distributed VMs, multi-site with AVFS replicas
- Minimum VM sizes: Azure A3 or larger, AWS m3.large or larger

Migration procedure:

1. Full backup of Vault database and file store
2. Test migration in non-production environment
3. Upgrade in steps no larger than two major versions
4. Engage an Autodesk reseller/VAR for complex migrations

The ADMS Console migration wizard handles internal path updates. Full documentation: [Vault Admin Help — Migrate a Database](https://help.autodesk.com/cloudhelp/2025/ENU/Vault-Admin/files/GUID-7C6C668C-4AC1-4199-BE04-268E5F80B949.htm)

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3. FORMAT AND APPLICATION COMPATIBILITY

.iam / .ipt / .idw (Inventor)

Confirmed compatible. The Vault add-in communicates with the ADMS API over TCP/IP — server location (on-premise vs cloud) is transparent to the client. Existing file relationships, revision history, properties, and lifecycle states are fully preserved through the migration. Post-migration: each workstation must reconfigure Vault client to point to the new server address.

.dwg / .dxf (AutoCAD)

Confirmed compatible. DWG xref paths are resolved by Vault to the local workspace on checkout and re-mapped on check-in — identical behaviour on cloud and on-premise. DXF files are flat managed files with no dependency resolution required.

AutoCAD Plant 3D (.xml, lock files, piping, isos)

Compatible with known constraints — constraints are unchanged by cloud hosting.

The safe workflow (check-IN from Plant 3D to Vault; do not check-OUT back into Plant 3D) is a Plant 3D application behaviour, not a server architecture issue. Cloud hosting does not introduce this constraint — it exists on-premise too. Cloud backup improves the protection of Plant 3D XML project data.

Caveat: Plant 3D project.xml files that reference old server hostnames will need path remediation as part of migration. Lock files (.lck) should be cleared before migration — they are transient state.

PDF, STEP/IGES, IFC, Excel BOM exports

All confirmed compatible as flat managed files. Standard migration includes all. IFC and Excel BOM exports are stored as documents; cloud hosting has no impact on their management.

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4. RECOMMENDED MIGRATION SEQUENCE

Phase 0 — Pre-migration hygiene (2–4 weeks)

- Audit all files outside Vault on network shares; check in before migration
- Clear all checked-out files; resolve orphaned check-outs

- Clear Plant 3D lock files
- Verify and finalise Vault folder structure (reorganise before migration, not after)
- Document current Vault server version; determine upgrade path if needed
- Full Vault backup (database + file store) as pre-migration baseline

Phase 1 — Cloud environment provisioning (1–2 weeks)

- Provision cloud VM (AWS m5.large or better / Azure D-series equivalent), Windows Server, SQL Server
- Install Vault Professional Server; configure VPN tunnel and firewall rules (TCP 25734, 25735)

Phase 2 — Test migration (1–2 weeks)

- Restore database backup to cloud SQL; copy file store; run ADMS Console migration wizard
- Test all client workflows (NOT Plant 3D check-out — check-in only)
- Measure performance baselines from office network over VPN

Phase 3 — Cutover (planned downtime, target <4 hours)

- Final incremental backup/restore; update client configurations across all workstations

Phase 4 — Post-migration validation (1 week)

- Monitor logs; verify Inventor xref resolution; verify Plant 3D check-in; confirm backup automation

5. WHAT SUCCESS LOOKS LIKE — MEASURABLE OPERATIONAL TERMS

METRIC	TARGET
Vault check-in/check-out round-trip latency	≤ on-premise baseline (or within 10% with AVFS)
Inventor xref resolution time on file open	≤ on-premise baseline
Vault uptime	≥ 99.5% (cloud provider SLA-backed)
Backup restore test	Full restore validated within 30 days of go-live
Zero data loss	Full revision history, lifecycle states, and properties preserved
IT maintenance events per quarter	Reduced from on-premise baseline
Plant 3D check-in failures	Zero increase vs on-premise baseline
Remote worker access	Full Vault client functionality confirmed from off-site
Time to recover from server failure	Cloud snapshot restore < 4 hours

6. COMPARABLE ORGANISATIONS

- **Shimano Batam (manufacturing):** Migrated Vault Professional to cloud for Japan HQ collaboration ([Autodesk case study](https://www.autodesk.com/ca-fr/support/partners/success-stories/pt-shimano-batam-advantages-in-collaborating-with-japan-hq-in-migrating-of-vault-professional/8453))
- **AU 2023 "Vault Shining on Clouded Days":** Engineering firm migrated Vault Professional to AWS in 2022–2023; session covered performance evaluation
- **IMAGINiT Managed Services client base:** Significant installed base of engineering organisations running Vault cloud under managed service agreements

SUMMARY OF ADVOCATE POSITION

The migration is:

1. **Technically sound** — Vault is the same software stack on a VM as on physical hardware
2. **Officially supported** — Autodesk documents cloud VM deployment topologies and migration procedures
3. **Operationally superior** — backup/DR, scalability, IT overhead
4. **Well-precedented** — engineering organisations of comparable profile have done this
5. **Safely sequenced** — full data preservation, validated test run, defined rollback before cutover
6. **Plant 3D caveat is neutral** — the known constraint is unchanged by cloud hosting; cloud improves backup

The strongest DA objection will be latency for large CAD file operations. The counter is AVFS replication — a local file cache at the office while Vault Server lives in the cloud. This is Vault Professional's canonical multi-site architecture.

Sources:

- [Vault Upgrade/Migration Strategy — Autodesk](https://autodesk.com/support/technical/article/Strategy-and-path-for-upgrading-Vault)
- [Migrate a Database — Vault Admin Help 2025](https://help.autodesk.com/cloudhelp/2025/ENU/Vault-Admin/files/GUID-7C6C668C-4AC1-4199-BE04-268E5F80B949.htm)
- [Vault Shining on Clouded Days 2023 — Autodesk University](https://www.autodesk.com/autodesk-university/class/Vault-Shining-on-Clouded-Days-2023)
- [Vault Managed Services — IMAGINiT](https://resources.imaginit.com/data-management/vault-managed-services-simplifying-autodesk-vault-operations-in-the-cloud)
- [SkyVault class handout MFG229048 — Autodesk University](https://static.au-uw2-stg.autodesk.com/Class_Handout_MFG229048_SkyVault_The_Quantum_of_Collaboration_Irvin_Hayes.pdf)
- [Shimano Batam Case Study — Autodesk](https://www.autodesk.com/ca-fr/support/partners/success-stories/pt-shimano-batam-advantages-in-collaborating-with-japan-hq-in-migrating-of-vault-professional/8453)
- [Plant 3D and Vault Workshare — AU 2015](https://www.autodesk.com/autodesk-university/class/Workshare-AutoCAD-Plant-3D-and-Vault-Professional-Setup-Synchronization-and-Replication-2015)

DEVIL'S ADVOCATE AGENT

DA Output — Strongest Case AGAINST

Full devil's advocate position. All failure modes and risks.

DA OUTPUT: Devil's Advocate — Autodesk Vault Professional Cloud Migration

MAD Cycle | Topic: Vault On-Premise to Cloud Migration

Role: Devil's Advocate (DA) — Find Every Way This Fails

Run date: October 2026

FRAMING NOTE

The AA gives you reasons to migrate. My job is to ensure you migrate knowing exactly what can kill you. The failure modes below are grounded in how Vault, Inventor, Plant 3D, and AutoCAD actually behave under the conditions this environment creates. Structured from most to least catastrophic.

SECTION 1: PLANT 3D — HIGHEST RISK

1.1 Check-In Direction Reversal Triggered by Path/UX Change

Scenario: A user, after migration, navigates to a Plant 3D drawing from Vault Explorer or the Vault web client and clicks "Get Latest." This appears identical to the pre-migration workflow. However, the file is now opened directly from the Vault-controlled working folder on the cloud-hosted path — not from within Plant 3D's project environment. The user makes edits and attempts to sync back. The check-in direction is effectively reversed. Plant 3D's project linkage breaks silently.

Why cloud-specific: On-premise, users had muscle memory tied to a specific folder structure. Cloud migration changes the displayed server path, how mapped drives appear, and how the Vault working folder is presented. Users who previously had `V:\` now have a UNC path or a different letter. This is enough to cause someone to use the wrong workflow — not from negligence, but because the visual cues changed.

Additional trigger: If the Vault working folder path changes as part of migration (standard practice), any existing project.xml referencing the old path becomes invalid. A user opening that project.xml sees an error and may attempt to "fix" it by pulling files from Vault directly — reversing the direction.

Rating: CRITICAL

IT test would catch this: No — IT tests whether files transfer, not whether workflow direction is preserved

Required test: A Plant 3D operator must deliberately attempt the wrong workflow (get from Vault, open in Plant 3D) on a migrated project and document exactly what breaks. This test must run before go-live.

1.2 Project.xml Hostname Reference Invalidation

Scenario: Every Plant 3D project contains a project.xml storing absolute references to project paths, including the SQL Express data source connection string which references the server by hostname. If the cloud Vault server hostname changes from (e.g.) `ENGSERVER01` to `vault.company.cloud.com`, every project.xml in every Plant 3D project may need updating. This is not a Vault migration step — it is a Plant 3D configuration step that Vault migration tooling does not touch.

What breaks: When a user opens a Plant 3D project and project.xml still references the old server name, Plant 3D throws a connection error. The project opens in degraded mode. Pipe specs, line numbers, and P&ID data are inaccessible.

Isometric generation fails. This is total loss of Plant 3D functionality.

Compounding factor: If the Plant 3D data source SQL instance did not move with Vault, Plant 3D must now reach both a cloud Vault server AND an on-premise SQL instance — a split architecture with its own failure modes.

Rating: CRITICAL

IT test would catch this: Unlikely — IT checks "can I open the file," not "can Plant 3D reach all dependent services"

Required test: Open every Plant 3D project on the migrated system with a Plant 3D operator. Confirm Data Manager connects, line list loads, isometric generation runs to completion. Document every project.xml path edit required.

1.3 Lock File Race Condition Under Cloud Latency

Scenario: Plant 3D uses `.plck` lock files to prevent concurrent editing. On LAN, lock file operations are near-instantaneous. At 50ms cloud latency:

- User A opens a P&ID drawing. A `.plck` file is written.
- User B attempts to open the same drawing 200ms later. The lock file write has not propagated. User B does not see the lock. Both users have the file open simultaneously.
- Both check in. One version silently overwrites the other. No conflict warning appears.

Why this is worse than it sounds: Plant 3D drawings contain embedded database references, pipe routing data, and xref chains. A silent overwrite corrupts the relational data linking P&IDs to isometrics to line lists. This corruption may not be discovered until an isometric is generated and routing doesn't match the P&ID.

Rating: CRITICAL

IT test would catch this: No — concurrent timing-sensitive testing is almost never in IT migration test plans

Required test: Two users simultaneously open the same Plant 3D drawing on the migrated system and both attempt to edit and check in. Repeat at peak network load. Verify lock file behaviour under each condition.

1.4 Isometric Generation Output Path Mismatch

Scenario: If the Vault working folder path changed post-migration, isogen may write output files to a location outside the Vault working folder. These files exist on the local machine, are never checked into Vault, and are invisible to other users. The user sees no error. Vault shows no new isometric. The next user to request that isometric re-runs generation from scratch.

Secondary scenario: Isogen configuration files (.ini files, spec files) reference absolute paths. If those paths changed as part of working folder migration, isogen produces incorrect output silently — incorrect line styles, incorrect spec callouts.

Rating: SIGNIFICANT

IT test would catch this: No — IT tests whether isogen runs, not whether output lands in the correct Vault location with correct content

Required test: Run isogen on a known P&ID on the migrated system. Confirm output appears in Vault under the correct revision. Compare output content against a known-good reference from the on-premise system. Inspect all isogen .ini files for path references before migration.

1.5 Pipe Spec and Catalog Path Invalidation

Scenario: Plant 3D pipe specs are stored in a shared catalog location referenced by the project. If the catalog was on the old server (e.g., `\\ENGSERVER01\PlantCatalog\`) and that path is not replicated to the cloud, every pipe spec lookup fails silently. Users can still draw pipe, but spec validation no longer runs. Non-standard components can be placed without warning. This defect can persist undetected through an entire project phase.

Rating: SIGNIFICANT

IT test would catch this: Unlikely

Required test: Attempt to place a restricted-spec pipe component on the migrated system. Confirm the spec validation fires correctly. Confirm the catalog path in project.xml points to an accessible location.

SECTION 2: INVENTOR SPECIFICS

2.1 Assembly Resolution Latency — The Arithmetic Problem

Scenario: Inventor resolves assembly files by loading each child component individually on open — a largely sequential process with no parallelisation.

Realistic worst-case calculation:

- Cloud round-trip: 60ms (corporate proxy + firewall + cloud routing — conservative)
- Files per assembly open: 350 (300 parts + 50 xref/reference operations)
- Single user, uncongested: 350 × 60ms = **21 seconds of pure network wait added per open**
- Pre-migration assembly open time for complex assembly: ~60 seconds
- Post-migration single user: **~81 seconds**

Under concurrent deadline load (10 users simultaneously):

- Effective latency per operation: 120ms (contention)
- 350 × 120ms = **42 seconds of network wait**
- Post-migration per user: **~102 seconds per assembly open**

This is not a minor inconvenience. Engineers who wait 100 seconds to open an assembly stop opening assemblies. They work from last-known local copies. Those copies diverge from Vault. Revision control is bypassed in practice even if it works technically.

Rating: CRITICAL if effective latency exceeds 60ms; SIGNIFICANT at 30–60ms

IT test would catch this: No — IT measures "can the file be accessed," not "how long does an assembly actually take to open under load"

Required test: Measure actual round-trip latency from every workstation through the corporate proxy. Time the open of the three largest assemblies with 10 concurrent users simulated. Establish the acceptable threshold before declaring this a blocker.

2.2 iLogic Rules Referencing Vault Properties Returning Null

Scenario: iLogic rules that read Vault properties (revision, lifecycle state, custom properties) use the Vault API through the Vault add-in. If the cloud Vault server URL changes, rules instantiating `VaultConnection` or using `Vault.Application.ActiveVaultServer` may fail silently — returning null or stale data rather than throwing an error.

Specific failure: A rule reads the lifecycle state to determine whether to lock a dimension table. On cloud Vault, the property call returns null. The rule treats null as "not released." A released drawing has its locked dimensions silently unlocked. Edits are made. The drawing is issued with incorrect dimensions.

Rating: CRITICAL for drawings with iLogic-governed properties

IT test would catch this: No — IT does not test iLogic logic, only whether Inventor opens

Required test: Identify every iLogic rule that reads a Vault property. Run each rule on the migrated system. Verify returned values against the Vault database directly.

2.3 Vault Add-In Version Mismatch — Day-One Outage

Scenario: Cloud Vault may be provisioned at a newer version than currently on-premise. Vault client add-in for Inventor is version-locked to the server version. Cloud Vault 2025 + workstation Vault 2024 client add-in = connection refused. The user cannot access Vault from Inventor at all.

The update cascade: Updating the Vault client add-in requires updating Autodesk Desktop Connector or reinstalling Inventor components. In a corporate environment with limited admin rights, this requires IT involvement for every affected workstation. If this happens at go-live, all users are offline simultaneously.

Rating: CRITICAL

IT test would catch this: Only if tested on every individual workstation — rarely done; IT tests a reference machine and assumes uniformity

Required test: Document the exact Vault client add-in version on every workstation. Cross-reference against Autodesk's compatibility matrix for the cloud Vault server version. Confirm match before go-live.

2.4 In-Progress Checked-Out Files Orphaned by Working Folder Change

Scenario: If the Vault working folder path changes as part of migration, files already checked out to the old path are orphaned. An engineer with a 200-part assembly checked out mid-design cannot check it back in. Vault cannot find the files.

Worst case: The engineer manually moves files and re-points Inventor's working folder. During this process, Inventor may re-resolve the assembly against Vault and pull older versions of components, overwriting in-progress work.

Rating: SIGNIFICANT

IT test would catch this: No — test cases use clean checkouts, not mid-work in-progress states

Required test: Before migration, audit all currently checked-out files across all users. Define the explicit procedure for each engineer to resolve checked-out state. Make this a hard migration prerequisite.

2.5 Pack and Go Performance

Scenario: Pack and Go on cloud Vault must traverse the full assembly tree, downloading each file to the local cache, then re-uploading copies as a new Vault package. For a 300-part, 500MB assembly: 500MB download + 500MB upload = 1GB transfer. At 50Mbps effective throughput, this takes 160+ seconds. Previously on LAN: under 10 seconds.

Rating: MINOR to SIGNIFICANT depending on usage frequency

IT test would catch this: No

Required test: Run Pack and Go on the largest assembly in the system on the cloud environment. Measure completion time.

SECTION 3: AUTOCAD SPECIFICS

3.1 Xref Resolution After Working Folder Change

Scenario: Absolute path xrefs hardcoded to `\\ENGSERVER01\drawings\` break entirely when the server changes. Relative path xrefs that traverse up directories fail if the working folder root changes.

Silent failure: AutoCAD opens the drawing. Xref is listed as "Not Found" in the xref palette. The drawing appears blank or partial. Under deadline pressure, the user proceeds without the xref, makes changes, checks in. The revision now has missing references — invisible in Vault, visible only when someone opens the drawing.

Rating: SIGNIFICANT

IT test would catch this: Unlikely — IT opens a drawing and checks whether it "opens," not whether all xrefs resolved

Required test: Open every drawing with xref checking enabled on the migrated system. Generate an xref audit report before and after migration. Any discrepancy is a migration defect.

3.2 Sheet Set Files (.dst) — Paths Break Silently

Scenario: Sheet set files reference individual drawings by absolute path. A sheet set for a project may reference 50 drawings at `\\ENGSERVER01\projects\PJ00193\drawings\`. Post-migration, these paths no longer resolve. The sheet set opens but shows all sheets as missing.

This is often not discovered until a plot run: Migration appears successful for weeks until someone attempts a batch plot for a submission. At that point, all sheet set paths fail and the batch plot produces nothing.

Rating: SIGNIFICANT

IT test would catch this: No — IT does not typically test sheet set batch plotting

Required test: Identify all .dst files in the Vault system. Perform a batch plot of each sheet set on the migrated system. Verify output matches pre-migration reference plots.

3.3 AutoCAD Vault Add-In Version Compatibility

Same version-lock issue as Inventor. AutoCAD 2023 workstations + Vault 2025 server = add-in will not connect. Users lose check-in/check-out from within AutoCAD and must use Vault Explorer as a fallback — slower and more error-prone.

Rating: SIGNIFICANT

Required test: Same explicit version matrix verification as Inventor, for every workstation.

SECTION 4: PDF AND DOCUMENT FORMATS

4.1 Manually-Attached PDF Metadata Loss

Scenario: PDFs manually attached as standalone Vault documents (vendor datasheets, transmittals) may be migrated without their Vault property metadata — specifically "Issued To," "Transmittal Number," and custom category fields. Post-migration, searches for "all documents issued to Contractor X" return an incomplete result set. Historical transmittal integrity is compromised.

Rating: SIGNIFICANT — compliance risk if project is subject to document control requirements

IT test would catch this: No — IT verifies file count, not metadata completeness

Required test: Select 50 manually-attached PDFs. After migration, verify all Vault properties match pre-migration values exactly. Use a scripted comparison, not a manual spot-check.

4.2 DWF/DWFx Visualisation Attachments Dropped

Scenario: If migration configuration drops DWF/DWFx visualisation attachments while preserving parent drawings, users viewing drawings in Vault's web viewer see an outdated or blank visualisation — even though the drawing itself is current. Engineers using web review workflows are effectively blind.

Rating: MINOR to SIGNIFICANT depending on reliance on web review

IT test would catch this: Unlikely

Required test: After migration, open 20 random drawings in Vault's web viewer. Confirm the visualisation is current.

SECTION 5: OTHER FORMATS

5.1 IFC/STEP/IGES Output Attachment Links Broken

Scenario: IFC, STEP, and IGES files stored as output attachments linked to native source files must preserve the parent-child link through migration. If migration breaks these links, Vault no longer knows which native file produced which IFC. Revision history of issued IFC files becomes untraceable.

Rating: SIGNIFICANT

IT test would catch this: No

Required test: Select 10 IFC files that are output attachments. After migration, verify the parent-child link is intact and lifecycle state is unchanged.

5.2 Excel BOM Item Numbers — If These Change, History Is Permanently Broken

Scenario: If the SQL database is rebuilt rather than migrated in-place and Vault item numbers change, every historical BOM export is misaligned with the Vault item master. Engineers comparing historical BOMs to the current Vault state get false differences. This is not recoverable — there is no automated way to reconcile historical Excel exports with new numbering.

Rating: CRITICAL if item numbers change — requires explicit pre/post confirmation

IT test would catch this: No

Required test: Export the item master for three projects before migration. After migration, export again. Compare item numbers row-by-row. Any discrepancy is a critical migration defect.

SECTION 6: DATA INTEGRITY

6.1 SQL Relationship Graph — Staged Migration Orphaning

Scenario: If the Vault file store and SQL database are migrated at slightly different points in time (common in staged migrations), files checked in during the window may be in the file store but not in SQL, or referenced in SQL but absent from the file store. These orphaned records surface weeks later when someone tries to open an old revision.

Rating: CRITICAL

IT test would catch this: No — IT runs a SQL restore, confirms it restores without errors, and marks migration complete

Required test: Run Vault's built-in consistency check tool (AuditVault or equivalent) after migration. Check for orphaned records, missing files, broken parent-child links. Document output. Resolve all errors before go-live.

6.2 SQL Config Still Pointing to Old Server Hostname

Scenario: Vault's SQL database stores the server hostname in configuration tables. If the cloud server has a different hostname, clients attempting to connect via the Vault add-in may silently be redirected to the old on-premise server. Users can log into the Vault web interface but cannot check in or check out from within Inventor. The add-in falls back to read-only mode. Users assume Vault is "slow" rather than broken.

Rating: CRITICAL

IT test would catch this: Only if the test is run from a client machine after Vault server config is updated AND someone specifically checks add-in behaviour

Required test: After migration, verify Vault server hostname in SQL configuration tables explicitly. Then connect from a representative client machine and confirm check-in and check-out work from within Inventor, AutoCAD, and Plant 3D.

6.3 Missing Binary Files for Historical Revisions

Scenario: During migration, if the file store migration is interrupted and resumed, some intermediate revision versions may have their SQL pointer present but binary file missing. These are invisible in normal use — Vault shows the revision in history but returns an error when a user attempts to open that specific old version. This surfaces at audit time: "produce revision 3 of drawing 47A-P-001." Vault shows it exists. Vault cannot open it. Compliance failure.

Rating: CRITICAL

IT test would catch this: No — IT does not open old revisions during testing

Required test: For a sample of 100 files across all projects, open every revision in Vault history. Confirm each version opens successfully. Automate with a script if possible.

SECTION 7: PERFORMANCE AND LATENCY

7.1 Concurrent Peak Load — Full Calculation

CONDITION	LATENCY/OP	FILES/OPEN	ADDED WAIT	TOTAL OPEN TIME
LAN baseline	1ms	350	0.35s	~60s
Cloud, single user	60ms	350	21s	~81s
Cloud, 10 concurrent users	120ms	350	42s	~102s

At 102 seconds per assembly open under deadline conditions, engineers stop using Vault workflows and revert to local copies. Revision control fails in practice.

Rating: CRITICAL if latency ≥60ms; SIGNIFICANT at 30–60ms

IT test would catch this: No

Required test: Measure actual round-trip latency from every workstation through the corporate proxy to the cloud server. Time the three largest assemblies with 10 concurrent users simulated. Establish the acceptable threshold before go-live.

7.2 Corporate Proxy SSL Inspection Overhead

Scenario: Vault's cloud connection may route through a corporate proxy performing SSL inspection. SSL inspection adds 20–40ms per connection establishment. Vault makes many short connections (one per file operation) — this overhead compounds multiplicatively. An apparent 30ms round-trip becomes an effective 70ms per file operation.

Rating: SIGNIFICANT

IT test would catch this: Unlikely — proxy overhead is rarely measured during application testing

Required test: Test Vault connection latency with and without proxy to isolate overhead. Request a proxy bypass for Vault traffic if overhead is significant and confirm security policy permits it.

SECTION 8: ROLLBACK AND PARALLEL RUN

8.1 Two Weeks of Data Loss If Rollback Required

Scenario: If a critical Plant 3D workflow failure is discovered two weeks post-go-live, rollback means reverting to the on-premise server snapshot. Any data created in those two weeks exists only in the cloud Vault. Rolling back means either accepting that loss, or attempting a forward-migration from cloud back to on-premise — not a tested or documented procedure, especially if cloud Vault is a newer version.

Nobody has planned for option (b). The rollback plan everyone has documented says "go back to the snapshot." That is a two-week data loss event.

Rating: CRITICAL

IT test would catch this: The rollback is documented. It has almost certainly not been executed end-to-end.

Required test: Execute the rollback procedure in a test environment before go-live. Not document it — run it. Confirm every file is recoverable. Define maximum acceptable data loss and get management sign-off before migration.

8.2 Parallel Run Data Divergence

Scenario: If both on-premise and cloud Vault are live simultaneously, check-ins to one are invisible to the other. After 48 hours, the two systems diverge. Users whose Vault client still points to the old server (because IT missed reconfiguring their machine) create a split revision history. When on-premise is decommissioned, those revisions are lost.

Rating: SIGNIFICANT

Required test: Set the on-premise Vault to read-only immediately at go-live. Enforce this technically, not just by policy.

SECTION 9: TEST CASE ADEQUACY

The Single Most Important Question

Were the IT-written test cases created with input from the engineers who use Plant 3D daily?

If not, these gaps are structurally absent from the test cases:

- The Plant 3D check-in direction constraint (IT does not know this constraint exists)
- Isometric generation output path validation
- iLogic rule execution with Vault property reads
- Sheet set batch plotting
- Concurrent lock file behaviour under deadline load
- Opening historical revisions from Vault history

Required action: A Plant 3D operator and a senior Inventor user must review the test case list and sign off that their critical workflows are covered before testing begins.

Test data must be production-size: The largest assembly, the most complex Plant 3D project, the most heavily-xreffed AutoCAD drawing. No synthetic test data.

Failure condition testing is required: Mid-operation connection drop during check-in (Inventor), check-in (Plant 3D), isometric generation, and assembly open. Document recovery procedure for each.

SECTION 10: VERSION COMPATIBILITY

Autodesk's compatibility matrix for Vault + Inventor + AutoCAD + Plant 3D is narrow and unforgiving. The cloud Vault version must match the exact client add-in version on every workstation.

Plant 3D add-in specific risk: Plant 3D's Vault add-in is the least documented and least tested of the three integrations. Version compatibility between Plant 3D updates and specific Vault server versions has sparse public documentation. Confirm directly with Autodesk before migration.

Required action: Document the exact application version on every workstation. Cross-reference against Autodesk's official compatibility matrix. Confirm the cloud Vault server version matches. Get this in writing from the cloud provider.

SUMMARY: CRITICAL FAILURE MODES — GO/NO-GO CONDITIONS

#	FAILURE MODE	RATING	IT TEST CATCHES?
1.1	Plant 3D check-in direction reversal triggered by path/UX change	CRITICAL	No
1.2	Project.xml hostname reference invalidation	CRITICAL	No
1.3	Lock file race condition under cloud latency	CRITICAL	No
2.1	300-part assembly open time at concurrent cloud latency	CRITICAL/SIG	No
2.2	iLogic Vault property reads returning null	CRITICAL	No
2.3	Vault add-in version mismatch — day-one outage	CRITICAL	Per-machine only
5.2	SQL item number change breaking BOM history	CRITICAL (if occurs)	No
6.1	SQL relationship graph orphaning in staged migration	CRITICAL	No
6.2	SQL config still pointing to old server hostname	CRITICAL	Client-test only
6.3	Missing binary files for historical revisions	CRITICAL	No
7.1	Cumulative latency under concurrent peak load	CRITICAL (≥60ms)	No
8.1	Rollback data loss — untested procedure	CRITICAL	No
9.1	Test cases written without Plant 3D operator input	CRITICAL (meta)	N/A

DA CLOSING STATEMENT

This migration is not impossible. But if it proceeds without the tests named above — without Plant 3D operator sign-off on test cases, without a confirmed version compatibility matrix for every workstation, and without a tested rollback — it is a high-probability partial failure that will manifest as a Plant 3D workflow crisis within the first week of live operation.

The failure will not announce itself clearly. It will look like files being slow, then like files being wrong, then like an isometric that doesn't match a P&ID, at which point the root cause will be three migration steps back and very difficult to unwind.

A **conditional Go** is defensible. An **unconditional Go** is not.